

APPLIED DATA STRUCTURES & ALGORITHMS LAB (BCS-553)

B.Tech CSE 5th SEMESTER

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Lab Assignment 9: Constructors

Experiment No. 9.1 - Constructor with no parameters

Experiment Title:

Constructor with no parameters

Aim:

To set a field value using a constructor with no parameters.

Software / Tools Required:

Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

Theory:

A constructor is a special method that runs automatically when an object is created with new. Its name is the same as the class name and it has no return type. A constructor with parameters lets the caller supply the initial values. The keyword this refers to the current object, so this.width is the field while width alone is the parameter.

Algorithm / Procedure:

1. Define class Box with field width.
2. Write a no-argument constructor Box() that sets width = 10.
3. Create an object with new Box().
4. Print the width.

Program / Queries / Commands:

```
// Name: Mankirat Singh
// ID: BCS2024542
// Roll No: 24103032571
// Assignment 9, Task 1
// Question: Create class Box with int field width and a constructor
// with no parameters that sets width to 10. In main, create a Box
// and print "Width:" followed by the value.

class Box {
    int width;

    Box() {
        width = 10;
    }
}

public class Main {
    public static void main(String[] args) {
        Box b = new Box();
```

```
        System.out.println("Width: " + b.width);
    }
}
```

Input:

No input; the constructor sets width = 10.

Output:

```
Width: 10
```

Time / Space Complexity: Time O(1); Space O(1).

Result:

The program was compiled and executed successfully. The constructor initialised width to 10 when the object was created.

Viva-Voce Questions:

- What is a constructor?
- Does a constructor have a return type?
- When does a constructor run?

Experiment No. 9.2 - Constructor with a parameter

Experiment Title:

Constructor with a parameter

Aim:

To set a field using a parameterised constructor and the this keyword.

Software / Tools Required:

Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

Theory:

A constructor is a special method that runs automatically when an object is created with new. Its name is the same as the class name and it has no return type. A constructor with parameters lets the caller supply the initial values. The keyword this refers to the current object, so this.width is the field while width alone is the parameter.

Algorithm / Procedure:

1. Define class Box with field width.
2. Write constructor Box(int width) that sets this.width = width.
3. Create an object with new Box(25).
4. Print the width.

Program / Queries / Commands:

```
// Name: Mankirat Singh
// ID: BCS2024542
// Roll No: 24103032571
// Assignment 9, Task 2
// Question: Create class Box with int field width and a constructor
// that takes an int and sets width using this.width. In main, create
// a Box with 25 and print "Width:" followed by the value.

class Box {
    int width;

    Box(int width) {
        this.width = width;
    }
}

public class Main {
    public static void main(String[] args) {
        Box b = new Box(25);
        System.out.println("Width: " + b.width);
    }
}
```

Input:

Constructor argument: 25.

Output:

```
Width: 25
```

Time / Space Complexity: Time O(1); Space O(1).

Result:

The program was compiled and executed successfully. The parameterised constructor set width to 25.

Viva-Voce Questions:

- What does the this keyword refer to?
- Why is this.width = width needed?
- What is a parameterised constructor?

Experiment No. 9.3 - Class Counter

Experiment Title:

Class Counter

Aim:

To use a constructor to start a counter at 0 and a method to increase it.

Software / Tools Required:

Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

Theory:

A constructor is a special method that runs automatically when an object is created with new. Its name is the same as the class name and it has no return type. A constructor with parameters lets the caller supply the initial values. The keyword this refers to the current object, so this.width is the field while width alone is the parameter.

Algorithm / Procedure:

1. Define class Counter with field count.
2. Constructor sets count = 0.
3. increment() adds 1 to count.
4. Create one Counter, call increment() three times, print count.

Program / Queries / Commands:

```
// Name: Mankirat Singh
// ID: BCS2024542
// Roll No: 24103032571
// Assignment 9, Task 3
// Question: Create class Counter with int field count, set to 0 by
// its constructor, and method increment() that adds 1 to count. In
// main, create one Counter, call increment() three times, and print
// "Count:" followed by the value.

class Counter {
    int count;

    Counter() {
        count = 0;
    }

    void increment() {
        count = count + 1;
    }
}

public class Main {
    public static void main(String[] args) {
        Counter c = new Counter();
        c.increment();
        c.increment();
```

```
        c.increment();
        System.out.println("Count: " + c.count);
    }
}
```

Input:

No input; increment() is called 3 times.

Output:

```
Count: 3
```

Time / Space Complexity: Time O(1); Space O(1).

Result:

The program was compiled and executed successfully. The counter started at 0 and reached 3 after three increments.

Viva-Voce Questions:

- What value does count have after construction?
- What happens if no constructor is written?
- How does increment() change the object?